HEADROOM

A recovery support tool built from lived experience

know your headroom

Submitted to: Change Grow Live, Nottingham
Prepared by: David Brand, service user
Date: April 2026
Contact: 9rogr4mm3r@gmail.com

This document describes a digital recovery support tool currently in active use by the author, a CGL Nottingham service user, and proposes a collaborative relationship with CGL to develop, pilot, and evaluate it with a wider service user cohort.

1. Where This Came From

This application began in a keyworker session at CGL Nottingham. My keyworker introduced the stress-bucket model as a way of understanding why I drink — the idea that triggers and pressures accumulate over time, and that drinking happens when the bucket overflows. I rejected it at first, as I have rejected many suggestions over the years. It struck me as too simple.

I came back to it. What changed was not the model but my relationship with it. I am a software developer by profession. I began to think about what it would look like if the bucket were real — if I could see it filling, see what was going into it, and see the effect of deliberate action to empty it. Over several weeks, working with an AI development tool in my own time, I built that.

The application I built is now something I use every day. I open it in the morning. I acknowledge every persistent difficulty in my life — the ongoing legal proceedings around my child's care, the financial pressures, the distance from my partner in the Netherlands, the anxiety I carry as background noise. I log them all. The load indicator goes into the red. That is not a problem. That is honesty. The practice is then to spend the day working it back down through conscious, deliberate action — exercise, contact with people I trust, creative work, rest. The goal is to have it low by evening, which is when I am most at risk.

> *Stopping the drug is easy. It can be eliminated. What really needs to be eliminated is the person that the drug was used by.*

This framing — encountered by the author during recovery — captures what the application is for. It is not a drinking tracker. It is a tool for understanding and managing the psychological state of the person who drank.

2. The Therapeutic Model

The stress-bucket model, as introduced to me by my CGL keyworker, describes psychological capacity as a container. Stressors fill it. Coping actions empty it. When it overflows, breakdown or lapse behaviour occurs. The model is used widely in cognitive behavioural therapy and trauma-informed care, and resonates immediately with most people in recovery because it matches their experience — lapse rarely feels like a sudden event; it feels, in retrospect, like the inevitable result of a long build-up.

Headroom extends this model in three ways:

- **Persistent and situational load:** It distinguishes between

Some stressors are constants — ongoing circumstances that do not disappear overnight and cannot be resolved by a single coping action. Family court proceedings do not conclude because you meditated. Financial anxiety does not clear because you went for a walk. The application holds these constants in view, so that the user begins each day with an honest assessment of what they are carrying, rather than pretending the slate is clean.

- **accumulation visible over time:** It makes the

The history record is kept indefinitely. A lapse event that appears to come from nowhere in the moment is often preceded, when the data is examined, by a sustained period of high load extending back days or weeks. This is clinically significant. It makes the gradual build visible rather than invisible, and supports the development of self-awareness and pattern recognition.

- **action-oriented rather than passive:** It is

The application does not simply record. It prompts engagement. Every relief action logged — however small — is a deliberate act. The model teaches that small actions matter: a five-minute walk, a phone call, a meal cooked rather than skipped. The cumulative effect of small, conscious actions is the practice.

3. The Neuroscience

The application's central visual is a midline cross-section of the human brain — anatomically accurate, showing the structures most relevant to the recovery story. This is not decorative. It reflects a specific and clinically supported understanding of what happens during a lapse.

What the brain diagram shows

At low load, the prefrontal cortex — the front of the brain, responsible for impulse control, consequence evaluation, and long-term planning — is depicted as illuminated: golden, active, present. As load rises, it dims. Simultaneously, deeper structures begin to glow: the limbic system (emotional memory, threat response), the hypothalamus (stress hormone gateway), and at the highest levels, the basal ganglia — the seat of habituated, automatic behaviour.

This visual progression is neurologically accurate. There is substantial evidence from addiction neuroimaging that prefrontal cortex activity is suppressed during high craving states, while the mesolimbic dopamine pathway — the reward circuitry associated with habituated drug-seeking behaviour — shows elevated activity. The decision to use, in chronic addiction, is often not a decision in any conscious sense. It is the execution of a deeply learned automatic behaviour by brain systems that do not deliberate.

> *The experience of finding oneself in a pub or a shop with no felt sense of having decided to go is recognised in addiction medicine as automatic or pre-conscious behaviour. The prefrontal cortex has been sufficiently overwhelmed that the reward circuitry executes the habituated behaviour before conscious awareness catches up.*

This is what the author experiences. It is what many people in recovery from substance use disorder experience. Headroom is designed to interrupt this process — not in the moment of lapse, when intervention may be too late, but through the daily practice of maintaining prefrontal engagement. The morning inventory, the act of consciously acknowledging what one is carrying, the deliberate choice of relief actions — all of these are exercises in keeping the conscious, reasoning brain active and informed.

Headroom as a concept

The name Headroom was chosen deliberately. Headroom is the space between where you are and the threshold at which automatic behaviour takes over. The application does not aim to eliminate the load — life does not permit that. It aims to help the user maintain enough space between their current state and that threshold that conscious choice remains possible. That space is headroom. It is maintained by deliberate action. When it disappears, the drug fills it instead.

4. The Application

Headroom is a web-based application, accessible on any device with a browser, requiring no installation. It is password-protected for privacy. A read-only link can be shared with a supporter or keyworker, allowing them to see the user's current state and history in real time.

The Now view

The central screen shows the brain diagram, the current load percentage, and two columns: Load (things that are filling the capacity) and Relief (things that empty it). The user taps an item when something happens. Items have impact values — a conflict with a family member might add 20 points; a 30-minute walk might remove 20. Items are fully customisable. Users add their own triggers and relief actions with their own impact ratings, building a personal model of what affects them.

The History view

Every entry is stored with a full timestamp. The history chart shows the load level over any selected time range — 7 days, 30 days, 90 days, or all time. A shaded zone above 70% marks the critical region. The gradual accumulation that precedes a lapse — sometimes stretching back weeks — is visible here in a way that a daily mood diary or drinking tracker cannot show. This is one of the application's most clinically significant features.

The Patterns view

Aggregate statistics show which triggers appear most frequently, which relief actions are used most, and whether load tends to build on particular days or at particular hours. Over time this builds a picture of the user's personal load profile — information that is directly useful in keyworker sessions and in the development of personalised coping plans.

The supporter link

A cryptographically protected read-only link can be shared with anyone the user chooses — a partner, a family member, a keyworker. The recipient sees the current load level, the brain diagram, and the history chart. They cannot make changes. This feature supports the relational dimension of recovery: someone who cares about the user can see, in real time, when the load is building. They do not need to ask. The user does not need to explain.

5. Lived Experience as Design

The application was not designed by a clinician or a product team. It was designed by a person in recovery, for himself, and it reflects things that only lived experience makes visible.

The decision to keep all data permanently — rather than resetting daily — came from understanding that lapse does not come from nowhere. The decision to load the bucket in the morning rather than start from zero came from understanding that the triggers are always there; pretending otherwise is part of the problem. The supporter link came from a specific relationship in which the ability to see, without asking, matters. The brain diagram came from wanting to understand, not just manage — from finding neuroscience more motivating than metaphor.

None of these design decisions would have been obvious from outside the experience. All of them make the application more clinically useful than a generic mood tracker or coping skills app.

> *The stress-bucket model is taught in keyworker sessions across CGL's services. Headroom operationalises that model into a daily digital practice that service users can use independently, between sessions, to maintain the work that begins in the room.*

6. The Opportunity for CGL

Change Grow Live is one of the largest substance misuse services in the United Kingdom, working with over 100,000 people annually across more than 80 areas. The stress-bucket model is already embedded in the therapeutic approach used by CGL keyworkers. Headroom does not introduce a new model. It extends the existing one into the space between sessions.

The opportunity is this:

- A tool grounded in CGL's own clinical approach, built by a CGL service user from direct experience of that approach working
- A mechanism for extending keyworker support beyond the session — the service user's load and patterns are visible to the keyworker in real time if the user chooses to share them
- A source of clinically meaningful data — not symptom scores or self-reported units, but a continuous record of what specifically builds load for an individual and what reliably reduces it
- A basis for a clinical pilot that could generate publishable evidence on the utility of load-tracking tools in substance misuse recovery
- A product that, if validated through pilot, has commercial potential in adjacent markets — employee assistance programmes, NHS talking therapies, workplace wellbeing — that could generate sustainable revenue for CGL or its social enterprise arm

The application is already built and deployed. It requires no initial investment of development resource from CGL. What it requires is clinical input, access to a willing cohort of service

users for piloting, and a relationship — formal or informal — that gives the development process the clinical grounding it needs to become a genuinely evidence-based tool.

7. What We Are Proposing

This document is an opening of a conversation, not a formal proposal. The author is seeking to understand whether CGL Nottingham — and specifically the keyworker who first introduced the stress-bucket model — would be willing to engage with this project in any of the following ways:

Informal clinical input

The keyworker reviews the application and provides feedback on its clinical appropriateness, the accuracy of the therapeutic model as it is represented, and any safeguarding considerations that should be addressed before wider use.

Service user pilot

A small cohort of consenting CGL service users — perhaps five to ten people — is invited to use the application for a defined period (suggested: eight to twelve weeks) alongside their existing keyworker support. Feedback is gathered and used to refine the product. Outcome data (AUDIT scores, self-reported wellbeing, keyworker observation) is collected with appropriate consent to begin building an evidence base.

Clinical co-development

CGL contributes clinical expertise to the development of features planned for the next phase of the product: the morning check-in mode, the persistent/situational trigger distinction, the safeguarding layer, and the keyworker dashboard. This involvement would be credited and, if the product reaches commercialisation, compensated through a revenue-sharing or licensing arrangement to be agreed.

Endorsement

If CGL considers the approach sound, a letter of support from the organisation would significantly strengthen applications for innovation funding (Innovate UK, the Health Foundation, NHS England digital health accelerators) and would accelerate the product's path to clinical validation.

8. Safeguarding and Ethics

Any clinical use of a digital tool with people in recovery from substance misuse requires careful attention to safeguarding. The following considerations are acknowledged and will be addressed in the next phase of development:

- A crisis resource layer will be added: when a user's load is sustained above a critical threshold for a defined period, the application will surface relevant crisis contacts and support resources
- The supporter view does not enable the supporter to log events or affect the user's data in any way. The data belongs to the user
- All health data will be treated as special category data under UK GDPR, requiring explicit consent, clear privacy policy, lawful basis for processing, and the right to full data erasure
- The application will not be presented as a clinical intervention or a substitute for keyworker support; it is a self-management tool that complements existing care
- A Data Protection Impact Assessment will be conducted prior to any formal clinical deployment
- Clinical governance arrangements appropriate to the pilot structure will be agreed with CGL before any service users are recruited

9. Proposed Next Steps

The author proposes the following as an initial path forward:

- A meeting between the author and the keyworker at CGL Nottingham to discuss the application, its current state, and whether CGL wishes to engage further
- If appropriate, a demonstration of the application to CGL clinical or management staff
- Identification of the most appropriate governance and consent framework for a small pilot
- Agreement on the form of any ongoing involvement — clinical input, pilot facilitation, co-development, or endorsement

The author is under no illusion that this will be straightforward. Clinical pilot design, ethics approvals, data governance, and the coordination of a digital tool with an established service all take time and require institutional support. The intention is not to move fast but to move carefully and collaboratively, in a way that respects both the clinical environment and the people the service exists to help.

10. About the Author

David Brand is a data engineer working in UK higher education, with professional expertise in database architecture, data modelling, and software development. He is a CGL Nottingham service user, currently engaged with the service in relation to alcohol use disorder. He has been in active recovery and has used Headroom daily since its development began in early 2026.

He is a single father, with a younger child currently in the care of the local authority, working toward reunification. He has a long-standing interest in the neuroscience of behaviour and motivation, and holds the view — informed by both his professional background and his lived experience — that the most useful tools in recovery are those that make invisible processes visible and give the person in recovery genuine agency over their own state.

He built this application because nothing like it existed and because the model it encodes is the one that has worked for him. He is sharing it because he believes it could work for others, and because the organisation that gave him the model deserves to know what was made from it.

Contact: 9rogr4mm3r@gmail.com

Live application: https://trigger-level.onrender.com

This document accompanies a separate technical design document: Headroom — Product Design Document (v0.1, April 2026)